

HTML Assignment

Topic 1: Semantic Tags

Q1. What are semantic tags in HTML?

A: Semantic tags are HTML tags that have a clear meaning, like `<header>`, `<footer>`, `<article>`.

Q2. Why should we use semantic tags?

A: They make the code easy to read, help SEO, and improve accessibility.

Q3. Give 3 examples of semantic tags.

A: `<header>`, `<nav>`, `<section>`.

Q4. What is the difference between `<div>` and semantic tags?

A: `<div>` is just a container with no meaning, but semantic tags describe the content.

Topic 2: Attributes

Q5. What are attributes in HTML?

A: Attributes give extra information to an element, like `href` in `<a>`.

Q6. What is the difference between `id` and `class` attributes?

A: `id` is unique for one element, `class` can be used for many elements.

Q7. What does the `alt` attribute do in images?

A: It shows text if the image cannot load and helps screen readers.

Q8. What is the use of the `title` attribute?

A: It shows extra info when we hover over the element.

Topic 3: HTML Elements (4 Q&A)

Q9. What are HTML elements?

A: An HTML element is everything from the start tag to the end tag.

Q10. What is the difference between block and inline elements?

A: Block elements take full width like `<div>`, inline elements take only needed space like `<span>`.

Q11. What is the difference between `<ol>` and `<ul>`?

A: `<ol>` is an ordered list with numbers, `<ul>` is an unordered list with bullets.

Q12. What is a self-closing element? Give an example.

A: An element that does not need a closing tag, like `<br />` or `<img />`.

Topic 4: Forms and Inputs

Q13. What is the use of forms in HTML?

A: Forms are used to collect user input.

Q14. Name some input types in HTML.

A: Text, password, email, checkbox, radio, file.

Q15. What does the placeholder attribute do in inputs?

A: It shows a hint text inside the input box.

Q16. What is the difference between GET and POST method in forms?

A: GET sends data in the URL, POST sends data securely in the body.

Topic 5: Media

Q17. How do we add an image in HTML?

A: Using ``.

Q18. How do we add a video in HTML?

A: Using `<video src="video.mp4" controls></video>`.

Q19. How do we add audio in HTML?

A: Using `<audio src="song.mp3" controls></audio>`.

Q20. What is the difference between `<iframe>` and `<video>`?

A: `<iframe>` is used to embed other webpages or YouTube videos, `<video>` is for direct video files.

CSS Assignment

Topic 1: Selectors

Q1. What are CSS selectors?

A: Selectors are used to target HTML elements and apply styles.

Q2. What is the difference between id selector and class selector?

A: id selector uses # and is unique, class selector uses . and can be used many times.

Q3. What is the universal selector in CSS?

A: The universal selector is * and it selects all elements.

Topic 2: Box Model

Q4. What is the CSS box model?

A: The box model is how every element is shown with content, padding, border, and margin.

Q5. What is the difference between margin and padding?

A: Margin is space outside the border, padding is space inside the border.

Q6. What does border-box do in CSS?

A: It makes width and height include padding and border.

Topic 3: Positioning and Layout

Q7. What are the types of CSS positioning?

A: Static, relative, absolute, fixed, sticky.

Q8. What is the difference between relative and absolute positioning?

A: Relative moves an element based on its normal position, absolute moves it based on the nearest positioned parent.

Q9. What is the difference between flexbox and grid?

A: Flexbox is for one direction (row or column), grid is for two directions (rows and columns).

Topic 4: Responsive Design

Q10. What is responsive design?

A: Responsive design makes websites look good on all devices.

Q11. What are media queries in CSS?

A: Media queries are used to apply styles based on screen size or device type.

Q12. What is the difference between em, rem, and % units?

A: **em** is relative to parent, **rem** is relative to root, **%** is relative to container size.

Topic 5: Styling s

Q13. How do we change the text color in CSS?

A: Using `color: red;`

Q14. How do we change the background color in CSS?

A: Using `background-color: blue;`

Q15. How do we change font size in CSS?

A: Using `font-size: 16px;`

JavaScript Assignment

Section 1: DOM Manipulation

Q1. What is the DOM in JavaScript?

A: DOM means Document Object Model, it represents the webpage as a tree.

Q2. How do we select an element by id in JavaScript?

A: Using `document.getElementById("idName")`.

Q3. How do we select all elements by class name?

A: Using `document.getElementsByClassName("className")`.

Q4. How do we change the text of an element in JavaScript?

A: `document.getElementById("demo").innerText = "Hello";`

Q5. How do we create a new element with JavaScript?

A: Using `document.createElement("p")`.

Section 2: Control Flow

Q6. What is control flow in JavaScript?

A: Control flow decides the order in which code runs.

Q7. What is the difference between if and switch?

A: if checks conditions one by one, switch is used when checking many cases of one value.

Q8. What is a loop in JavaScript?

A: A loop repeats code until a condition is false.

Q9. Name 3 types of loops in JavaScript.

A: for, while, do...while.

Q10. What is the difference between break and continue?

A: break stops the loop, continue skips the current step and goes to the next.

Section 3: ES6 Features

Q11. What is ES6?

A: ES6 is a new version of JavaScript with modern features.

Q12. What is the difference between var, let, and const?

A: var is old, let is block-scoped, const is constant and cannot be changed.

Q13. What are arrow functions?

A: A shorter way to write functions using `() => {}`.

Q14. What is template literal in ES6?

A: A way to write strings with backticks ``` and `${}` for variables.

Q15. What is destructuring in ES6?

A: A way to unpack values from arrays or objects into variables.

Q16. What are default parameters in functions?

A: Parameters that have a value if no value is given.

Q17. What is the difference between map() and forEach()?

A: forEach just loops, map loops and returns a new array.

Q18. What are promises in JavaScript?

A: Promises are used to handle asynchronous tasks like fetching data.

Section 4: APIs

Q19. What is an API in simple words?

A: API is a way for programs to talk to each other.

Q20. What is the Fetch API in JavaScript?

A: Fetch API is used to get data from a server.

Q21. Write a simple example of fetch.

A: `fetch("url").then(res => res.json()).then(data => console.log(data));`

Q22. What is the difference between JSON and JavaScript object?

A: JSON is text format, JavaScript object is actual code object.

Q23. What does async and await do in JavaScript?

A: They make asynchronous code look simple like normal code.

Q24. What is the difference between GET and POST request in APIs?

A: GET is for reading data, POST is for sending data.

Q25. What happens if fetch request fails?

A: It throws an error which can be handled using `catch()`.